

# Feature Descriptions

## Customer Churn Dataset

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**Source Dataset:** WA\_Fn-UseC\_-Telco-Customer-Churn.csv

**Cleaned File:** clean\_dataset.csv

**Total Features:** 31

**Total Rows:** 7,043

## Preprocessing Notes

- customerID was dropped — it is a non-informative identifier.
- TotalCharges was converted from string to numeric; missing values were imputed with the column median.
- Churn was label-encoded: No → 0, Yes → 1.
- All remaining categorical columns were one-hot encoded using `pd.get_dummies(drop_first=True, dtype=int)`. Each encoded column contains binary values (0 or 1).

## Target Variable

| Column Name | Type             | Description                                                               |
|-------------|------------------|---------------------------------------------------------------------------|
| Churn       | Binary int (0/1) | Whether the customer left within the last month. 0 = Stayed, 1 = Churned. |

## Numeric Features

| Column Name    | Type             | Description                                                                                              |
|----------------|------------------|----------------------------------------------------------------------------------------------------------|
| SeniorCitizen  | Binary int (0/1) | Whether the customer is a senior citizen. 0 = No, 1 = Yes.                                               |
| Tenure         | Integer (0-72)   | Number of months the customer has been with the company.                                                 |
| MonthlyCharges | Float            | Amount charged to the customer each month (USD).                                                         |
| TotalCharges   | Float            | Total amount charged over the customer's entire tenure (USD). Missing values imputed with column median. |

## One-Hot Encoded Features

*All columns below are binary (0 = absent/No, 1 = present/Yes). The dropped reference category is shown in parentheses.*

## Demographics

| Column Name    | Original Column | Description                                 |
|----------------|-----------------|---------------------------------------------|
| gender_Male    | gender          | 1 if the customer is Male. (Ref: Female)    |
| Partner_Yes    | Partner         | 1 if the customer has a partner. (Ref: No)  |
| Dependents_Yes | Dependents      | 1 if the customer has dependents. (Ref: No) |

## Phone Services

| Column Name                    | Original Column | Description                                                            |
|--------------------------------|-----------------|------------------------------------------------------------------------|
| PhoneService_Yes               | PhoneService    | 1 if the customer has phone service. (Ref: No)                         |
| MultipleLines_No phone service | MultipleLines   | 1 if the customer has no phone service at all. (Ref: No — single line) |
| MultipleLines_Yes              | MultipleLines   | 1 if the customer has multiple phone lines. (Ref: No)                  |

## Internet Services

| Column Name                 | Original Column | Description                                             |
|-----------------------------|-----------------|---------------------------------------------------------|
| InternetService_Fiber optic | InternetService | 1 if the customer uses Fiber optic internet. (Ref: DSL) |
| InternetService_No          | InternetService | 1 if the customer has no internet service. (Ref: DSL)   |

## Online Security

| Column Name                        | Original Column | Description                                             |
|------------------------------------|-----------------|---------------------------------------------------------|
| OnlineSecurity_No internet service | OnlineSecurity  | 1 if the customer has no internet service. (Ref: No)    |
| OnlineSecurity_Yes                 | OnlineSecurity  | 1 if the customer has online security add-on. (Ref: No) |

## Online Backup

| Column Name                      | Original Column | Description                                           |
|----------------------------------|-----------------|-------------------------------------------------------|
| OnlineBackup_No internet service | OnlineBackup    | 1 if the customer has no internet service. (Ref: No)  |
| OnlineBackup_Yes                 | OnlineBackup    | 1 if the customer has online backup add-on. (Ref: No) |

## Device Protection

| Column Name                          | Original Column  | Description                                               |
|--------------------------------------|------------------|-----------------------------------------------------------|
| DeviceProtection_No internet service | DeviceProtection | 1 if the customer has no internet service. (Ref: No)      |
| DeviceProtection_Yes                 | DeviceProtection | 1 if the customer has device protection add-on. (Ref: No) |

## Tech Support

| Column Name                     | Original Column | Description                                          |
|---------------------------------|-----------------|------------------------------------------------------|
| TechSupport_No internet service | TechSupport     | 1 if the customer has no internet service. (Ref: No) |
| TechSupport_Yes                 | TechSupport     | 1 if the customer has tech support add-on. (Ref: No) |

## Streaming TV

| Column Name                     | Original Column | Description                                             |
|---------------------------------|-----------------|---------------------------------------------------------|
| StreamingTV_No internet service | StreamingTV     | 1 if the customer has no internet service. (Ref: No)    |
| StreamingTV_Yes                 | StreamingTV     | 1 if the customer subscribes to streaming TV. (Ref: No) |

## Streaming Movies

| Column Name                         | Original Column | Description                                                 |
|-------------------------------------|-----------------|-------------------------------------------------------------|
| StreamingMovies_No internet service | StreamingMovies | 1 if the customer has no internet service. (Ref: No)        |
| StreamingMovies_Yes                 | StreamingMovies | 1 if the customer subscribes to streaming movies. (Ref: No) |

## Contract

| Column Name       | Original Column | Description                                        |
|-------------------|-----------------|----------------------------------------------------|
| Contract_One year | Contract        | 1 if on a one-year contract. (Ref: Month-to-month) |
| Contract_Two year | Contract        | 1 if on a two-year contract. (Ref: Month-to-month) |

## Billing & Payment

| Column Name                           | Original Column  | Description                                                            |
|---------------------------------------|------------------|------------------------------------------------------------------------|
| PaperlessBilling_Yes                  | PaperlessBilling | 1 if the customer uses paperless billing. (Ref: No)                    |
| PaymentMethod_Credit card (automatic) | PaymentMethod    | 1 if the customer pays via automatic credit card. (Ref: Bank transfer) |
| PaymentMethod_Electronic check        | PaymentMethod    | 1 if the customer pays via electronic check. (Ref: Bank transfer)      |
| PaymentMethod_Mailed check            | PaymentMethod    | 1 if the customer pays via mailed check. (Ref: Bank transfer)          |